

KARTHIK PALYAKERE SURESH

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AI/ML Engineer and Researcher with 2+ years of hands-on experience building production-grade computer vision systems, generative AI pipelines, and multimodal deep learning models. Strong foundation in PyTorch, model optimization, and scalable MLOps workflows. Published researcher at ICCV 2025 with proven expertise spanning object detection, instance segmentation, foundation models (SAM2, CLIP, DINO), large language models, and retrieval-augmented generation. Comfortable working across the full stack from research prototyping to containerized deployment on cloud infrastructure.

CORE SKILLS

Programming: Python, C++, C, Bash, SQL, Git, Linux

Computer Vision: Object detection, instance segmentation, semantic segmentation, image classification, video perception, low-light vision, 2D/3D vision, image enhancement

Deep Learning & AI: CNNs (ResNet, MobileNet, EfficientNet, YOLO), Vision Transformers, CLIP, DINO, SAM2, Mask R-CNN, LSTMs, model fine-tuning, transfer learning

Generative AI & LLMs: Large Language Models, Retrieval-Augmented Generation (RAG), embeddings, vector databases, prompt engineering, synthetic data generation

Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face, OpenCV, Detectron2, ONNX Runtime, CUDA, NumPy, Albumentations, TensorRT, OpenVINO

MLOps & Deployment: Docker, FastAPI, AWS (EC2, S3, Lambda), CI/CD, ONNX, TensorRT, model monitoring, ETL pipelines, feature engineering

Languages: English (C2, Fluent) | German (B1, Actively Learning)

PROFESSIONAL EXPERIENCE

Working Student - Data Scientist | Merck Group KGaA, Darmstadt

March 2025 - Present

- Developed and evaluated computer vision models for high-resolution SEM and TEM microscopy data, fine-tuning instance segmentation and foundation models including Mask R-CNN, YOLOv11, SAM2, and SAM3
- Architected end-to-end ML pipelines on AWS infrastructure with containerized model serving via Docker, optimizing inference latency by 40% for large-scale, high-resolution imagery
- Engineered production-ready AI systems with CI/CD automation, reducing deployment cycles by 50% and enabling continuous model improvement through integrated feedback loops
- Designed synthetic data generation pipelines to improve model generalization and reduce overfitting by 35%, addressing limited labelled dataset constraints
- Prototyped and integrated LLM-based information extraction workflows using Retrieval-Augmented Generation (RAG), embeddings, and vector similarity search
- Implemented mixed-precision inference pipelines and benchmarked against FP32 baselines across latency, memory footprint, and throughput metrics
- Collaborated with interdisciplinary research and engineering teams in agile sprints, delivering features with clear documentation and test coverage

Research Assistant | DFKI - German Research Center for AI, Kaiserslautern

July 2023 - Feb 2025

- Designed and deployed a real-time multi-sensor perception system for robotic platforms, achieving 92% detection accuracy in autonomous scenarios
- Developed ROS2-based perception nodes for synchronized camera streams and object detection on Jetson embedded hardware, achieving sub-200ms inference latency
- Implemented low-light and degraded-vision enhancement pipelines, improving object detection mAP50 by 15% in challenging environments
- Optimized PyTorch models for production deployment focusing on inference performance, memory efficiency, and real-time constraints in resource-limited environments
- Built modular ML components and scalable data processing pipelines for structured and unstructured sensor data, ensuring robustness and reproducibility across distributed systems
- Developed monitoring and logging infrastructure for deployed AI services, enabling continuous performance tracking and automated alerting
- Published peer-reviewed research at ICCV 2025 demonstrating scalable, production-ready perception systems

PROJECTS

Real-Time Multi-Camera Perception System

- Built a real-time perception pipeline processing 4 camera streams at 30 FPS, integrating object detection, tracking (DeepSORT), and ROS2 communication for autonomous navigation
- Deployed on Jetson embedded hardware with GPU-optimized model serving and sub-200ms end-to-end latency

CLIP-DINO Hybrid Vision Transformer

- Developed a hybrid vision foundation model combining self-supervised (DINO) and language-aligned (CLIP) representations for zero-shot object recognition
- Deployed as a containerized inference service using FastAPI and Docker with automated CI/CD pipeline achieving sub-200ms response time

Adaptive Low-Light Detection and Segmentation

- Engineered a U-Net-based image enhancement pipeline integrated with object detection models (RTMDet-tiny, YOLOv3, DeeplabV3+), improving detection mAP50 by 15%
- Evaluated and benchmarked models on dark-image datasets, achieving 92% accuracy through systematic optimization

Retrieval-Augmented Generation (RAG) System

- Implemented a RAG prototype using LLM APIs, embeddings, and vector similarity search for domain-specific information retrieval
- Evaluated multiple retrieval strategies and optimized system architecture for answer relevance and consistency; deployed via FastAPI and Docker on AWS Lambda

CNN Quantization and Optimization Pipeline

- Applied Post-Training Quantization (PTQ) and Quantization-Aware Training (QAT) techniques with mixed-precision strategies (INT8/FP16)
- Achieved 2-3x inference speedup with controlled accuracy degradation, benchmarked against FP32 baselines

PUBLICATIONS

- **ICCV 2025** | TorchAdapt: Towards Light-Agnostic Real-Time Visual Perception
- **IJSRD, Vol.7, Issue 6, 2022** | Intellectual Interactive System
- Active open-source contributor in multimodal and computer vision research

EDUCATION

M.Sc. Computer Science (AI & Computer Vision Specialization) Oct 2022 - Expected June 2026

Technical University of Kaiserslautern, Germany

Relevant Coursework: Machine Learning Algorithms, Deep Learning, Image Processing, NLP, Web Mining

B.E. Computer Science Jul 2018 - Aug 2022

Nitte Meenakshi Institute of Technology, Bangalore, India

Relevant Coursework: Machine Learning, Artificial Intelligence

CERTIFICATIONS

- Oracle Cloud Infrastructure - Generative AI Professional (2025)
- Oracle Cloud Infrastructure - Data Science Professional (2025)